

Which Public Transport Modes Are Most Used in Victoria and Where Should Investment Be Focused?

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1 Executive Summary

This report examines the use of public transport in Victoria to identify the most popular modes and the areas where investment is likely to have the most impact. It combines a monthly statewide patronage dataset with the most recent FY2024–25 annual station-entry dataset for Melbourne metropolitan train stations. The analysis will compare demand between modes of transport and will highlight the stations with the highest levels of passenger activity. The aim is to enable practical investment decisions that enhance capacity, reliability and passenger experience in the areas of highest demand.

2 Introduction

Public transport is an important part of Victoria’s mobility network, connecting people to work, education, health care and other day-to-day activities. It is important to understand where the system is most heavily used as the population, commuting patterns and service expectations continue to change. This report looks at two related questions: which modes of public transport carry the most passengers in Victoria, and where investment should be directed in the metropolitan train network.

To answer these questions, the report uses two datasets. The first is a monthly record of boardings across the main public transport modes in Victoria, including metropolitan and regional train, tram, bus and coach services. The second is an annual record of passenger entries at metropolitan train stations in Melbourne based on the latest FY2024–25 data. These datasets allow a system-wide look at mode usage and a station-level look at where demand is concentrated.

The report is designed to identify patterns to inform infrastructure upgrades, service planning and network prioritisation. The report will assist in identifying where public transport investment is needed most by comparing patronage across modes and looking at the busiest train stations.

3 Methodology

3.1 Data source

This report uses two datasets sourced from the Victorian Government’s open data portal, both published by the Department of Transport and Planning and licensed under Creative Commons Attribution 4.0 (CC by 4.0). The first dataset contains monthly public transport boardings across six modes like metropolitan train, metropolitan tram, metropolitan bus, regional train, regional bus and regional coach, covering the period from 2018 to 2025. This dataset allows us to compare how much each mode is used across Victoria and how usage has changed over time. The second dataset contains annual station entries for all Melbourne metropolitan train stations broken down by day type. This dataset helps identify which stations carry the most passengers and where investment is most needed.

3.2 Data preparation

The data was loaded and processed in R using the `tidyverse` and `janitor` packages. For the monthly dataset, numeric columns were parsed to remove any formatting characters and the data was reshaped from wide to long format so that each row represents one mode in one month. For the station dataset, the data was filtered to the 2024-2025 financial year and numeric columns were cleaned in the same way. Both cleaned datasets were saved to the `data/processed/` folder to ensure the analysis is fully reproducible.

3.3 Analysis

To compare patronage across modes, total boardings were summed by mode across the full dataset. As shown in **?@tbl-mode-summary**, metropolitan train recorded the highest total boardings over the study period followed by metropolitan tram and metropolitan bus. Regional services made up a much smaller share of total patronage. To identify the busiest stations in the metropolitan train network, stations were ranked by total annual entries for 2024-2025. **?@tbl-top-stations** shows the top busiest stations with Flinders Street, Southern Cross and Melbourne Central recording significantly higher passenger volumes than all other stations. A bar chart summarising total boardings by mode is presented in **?@fig-mode-plot** to provide a visual comparison across the network.

```
# A tibble: 6 x 2
```

mode	total_boardings
<chr>	<dbl>
1 metropolitan_train	1366415976
2 metropolitan_tram	1128683201
3 metropolitan_bus	797593599
4 regional_train	146151590
5 regional_bus	92905490
6 regional_coach	8709111

```
# A tibble: 10 x 5
```

stop_name	pax_annual	pax_weekday	pax_saturday	pax_sunday
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 Flinders Street	19633300	60900	43150	31350
2 Southern Cross	14696050	47600	26500	19550
3 Melbourne Central	11876900	38300	20650	17250
4 Parliament	6344650	22350	6450	4750
5 Footscray	4383950	14000	8850	5800
6 Richmond	3712750	9850	13000	8850
7 Flagstaff	3445200	12350	2900	2200
8 Caulfield	3205100	9600	8250	5600
9 South Yarra	3199050	10350	5900	4100
10 Box Hill	2648350	8350	5350	4100

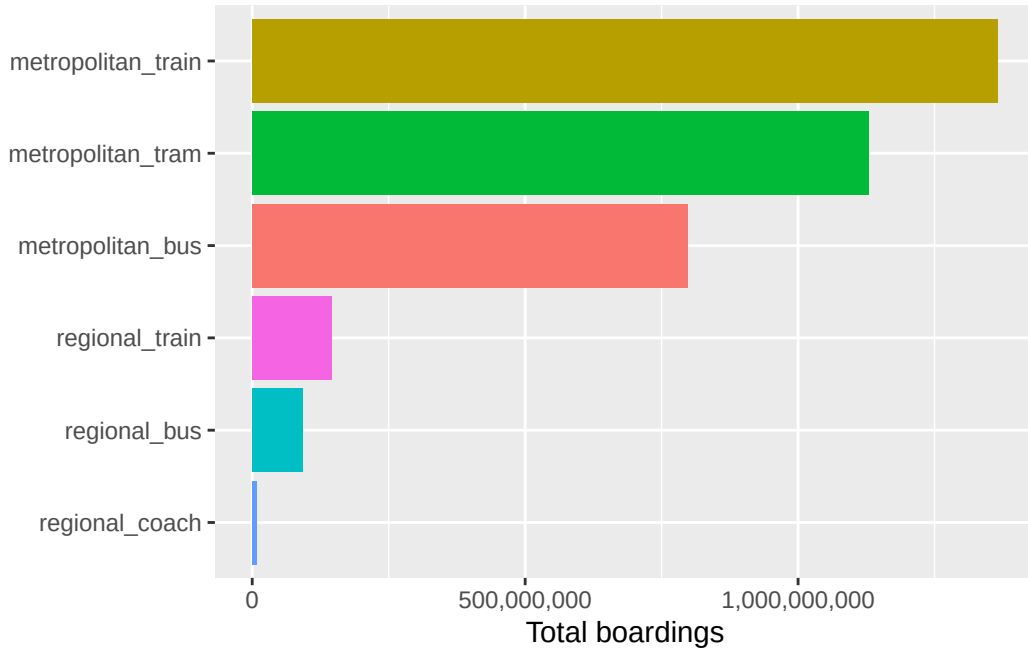


Figure 1: Total public transport boardings by mode across the monthly Victorian dataset.

4 Results

The results show clear differences in how Victorians use public transport across different modes. Metropolitan train recorded the highest total boardings over the study period, followed by metropolitan tram and metropolitan bus. Regional services and including regional train, regional bus and regional coach, made up a much smaller share of total patronage. As shown in Figure 2, all modes experienced a sharp drop in boardings around 2020 which lines up with COVID-19 lockdowns and restrictions in Victoria. After 2020, patronage gradually increased across all modes and though the recovery has been slow and uneven. Metropolitan train and tram showed stronger recovery compared to bus and regional services. As of 2024-25, none of the modes have returned to their pre-COVID levels which suggests that travel behaviour in Victoria has changed since the pandemic.

5 Discussion

The findings show that Victoria's public transport network is heavily concentrated around metropolitan train and tram services. This is not surprising given that Melbourne has a dense inner-city network and a large commuter population. However, the data also highlights that the network has not fully recovered from the impact of COVID-19 which raises questions about whether pre-pandemic service levels are still appropriate for current demand. The station-level data shows that a small number of stations, particularly Flinders Street, Southern Cross and Melbourne Central, handles a disproportionately large share of passenger entries. This puts pressure on these stations and the lines that connect to them.

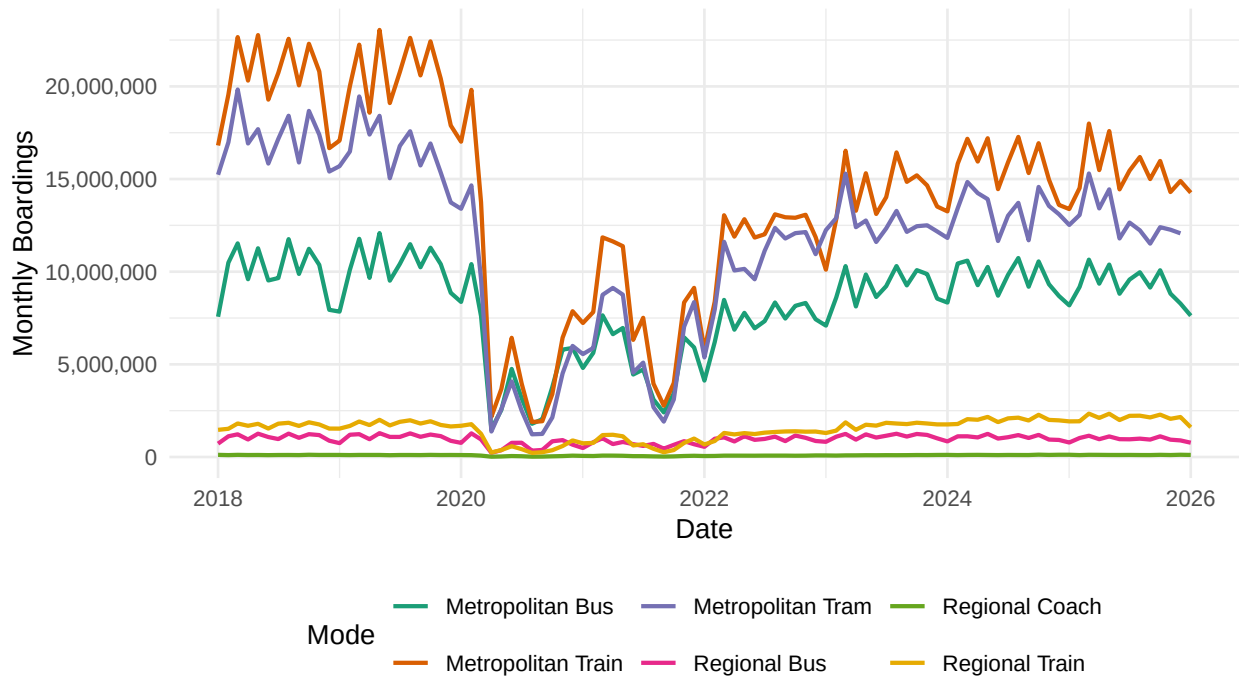


Figure 2: Monthly public transport boardings by mode in Victoria (2018–2025), showing the sharp decline during COVID-19 in 2020 and the subsequent recovery. Metropolitan train remains the dominant mode throughout the period.

6 Conclusion

Overall, metropolitan train is the most used public transport mode in Victoria, with tram coming in second. Demand is heavily concentrated at a few inner-city stations and patronage across all modes is still below pre-COVID levels as of 2024–2025. These findings suggest that investment should be focused on the modes and stations that carry the most passengers rather than spreading resources evenly across the network.

7 Recommendations

Based on the analysis, we recommend the following:

1. **Focus investment on high-demand stations** such as Flinders Street, Southern Cross and Melbourne Central, where passenger volumes are significantly higher than other stations on the network.
2. **Improve frequency and reliability on metropolitan train lines**, as this is the mode most used by Victorians and has the greatest potential impact on overall network performance.
3. **Track patronage recovery over time** across all modes to understand whether demand has permanently shifted or is still recovering before making major decisions about service cuts or expansions.

8 References

1. Monthly public transport patronage by mode. Victorian open data. [link](#).
2. Annual metropolitan train station patronage (station entries). Victorian open data. [link](#).